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Establishing weighted criteria for return to play after hamstring injury: a systematic review and methodological quality analysis

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Abstract

Introduction Hamstring injuries are the most prevalent in active individuals aged between 20 and 70 years, with a high recurrence rate of 57% during the first month. The lack of a clear consensus on the return to activity of these patients could lead to an increase in the incidence of relapses, as well as greater economic costs.

Methods A systematic review was carried out, following the PRISMA guidelines. The article search was conducted in the PubMed, Sportdiscuss, WOS and Scielo databases, using the terms "hamstring muscle" OR "biceps femoris" OR "semimembranosus" OR "semitendinosus" OR "hamstring tendon" OR "semitendinosus tendon" OR "semimembranosus tendon" OR "biceps femoris tendon" AND "return to sport" OR "return to play" OR "resumption of sporting activities" OR "return to sporting activities" OR "recreational activities resumption" OR "return to recreational activities".

Results 22 studies were finally included in the present review. The main variables analyzed were the presence of pain, the performance of functional tests, the measurement of flexibility and range of motion, the strength developed and the use of imaging tests.

Conclusion Pain measurement has been shown to be the most common criterion for determining return to activity, and its combination with other complementary measurements such as strength, flexibility and the ability to perform sport-specific movements at maximum speed is recommended.

Keywords Hamstring muscle, Return to sport, Physiotherapy, Recurrence

Introduction

Hamstring injury is the most prevalent type of injury in active people aged between 20 and 70 years [1]. In competitive sports, it is the main lesion in those sports where high-speed sprints are frequently performed, as well as jumps, kicks and explosive movements of the lower limb [2], together with changes of direction [3]. In football, it has an incidence of 4% to 30% of all injuries [4] and accounts for 25% of all sick leaves due to injury [5], with a recurrence rate of up to 68% [6] and an average absence time that varies between 14 and 20 days [7]. In addition, it is more frequent in men than in women [8], with an

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annual increase in incidence of 4% since 2001 in men´s elite football players [9]. It has also been associated in the literature with other sports, such as dancing [10] and athletics [11].

These injuries have a high recurrence rate, which can reach up to 63% two years after the injury, although 57% occur in the first month [12]. These relapses increase the rehabilitation period after the injury [13], which causes in the athlete an increase in frustration [14], decreased capacity to develop maximum strength [15], and reduced extensibility [16], resulting in an alteration of the structure and function of these muscles [17], together with a lower performance in sports practice [18]. Factors such as decreased muscle strength [19], inadequate rehabilitation [20], incomplete recovery [21] or a premature return to competition [22] constitute some of the risks of suffering a relapse after a muscle injury.

What criteria should be used to determine the return to play? In the scientific literature, there is no consensus that clearly answers this question [23]. To date, the criteria used for returning to activity have mainly been based on diagnostic imaging [24], in addition to a wide range of functional tests [25] and the absence of pain in specific tests for each sport [20]. Moreover, financial [26] and sports performance [18] reasons have gained relevance in elite sports, modifying the criteria for rehabilitation and return to activity [27], gaining ground over clinical decisions [28]. A consensus has not been established on the tests to predict recovery time [29], although factors that will influence them have been determined, such as the injury mechanism [30], the location of the injury [31], its extension seen in diagnostic imaging [32], and the degree of involvement of the proximal tendon [33]. Thus, not using adequate criteria for returning to physical activity could lead to new injuries, as well as greater economic costs.

The objective of this study is to analyze the scientific literature in order to establish defined criteria for clinical decision-making regarding the return to play following a hamstring muscle injury.

Materials and methods

Study design

This systematic review was conducted under the PRISMA [34, 34] regulations and registered in OSF under registration number https://doi.org/10.17605/OSF.IO/GSVE6.

Search strategy

The bibliographic search was carried out 20th April 2026, using the PubMed, SportDiscuss, Web of Science (WoS) and SciELO databases. To carry out the article search, the PRESS strategy [35] was followed. Thus, the terms and strategies used for the search in all databases cited were: ("hamstring muscle" OR "biceps femoris" OR "semimembranosus" OR "semitendinosus" OR "hamstring tendon" OR "semitendinosus tendon" OR "semimembranosus tendon" OR "biceps femoris tendon") AND ("return to sport" OR "return to play" OR "resumption of Sporting activities" OR "return to sporting activities" OR "recreational activities re-sumption" OR "return to recreational activities"). The MeSH terms mentioned above were included in all databases used in this review, following the structure described above. No filters were applied during the search, and all articles published up to the search date were considered. Grey literature, such as clinical trials, theses, and conference proceedings, was not included in this study.

Document selection

The inclusion criteria used are as follows: Studies dealing with patients who have suffered a hamstring injury, regardless of the grade of the injury and the level of physical activity performed. It must include at least one measurement variable for return to activity, regardless of whether it is compared to another variable. This integrative review incorporated a high-impact selection of scientific literature structured around three methodological pillars: randomized controlled trials to evaluate therapeutic interventions, which were appraised using the PEDro scale; observational longitudinal studies to establish prognostic indicators and biomechanical symmetry metrics, assessed via the STROBE checklist; and systematic reviews to synthesize international consensus, validated through the PRISMA checklist. Studies published in English, Spanish, French, Portuguese or Italian.

The exclusion criteria are patients who have undergone surgery. In addition, articles that are case reports or opinion pieces will also be excluded.

After carrying out the search, the obtained studies were classified independently and blindly by two researchers (D.P.C. and J.J.B.R.), using the Rayyan platform. This was conducted in two stages. Firstly, the titles and abstracts of the studies found were evaluated; then, the full text of said articles was evaluated. After this procedure, both researchers discussed those articles in which they presented different opinions. After the initial blinded screening, the inter-rater agreement (Cohen's kappa) was 0.81 and after the open discussion between both initial researcher as 0.96. In those articles in which the researchers did not reach a common opinion, a third reviewer (M.G.S.) decided on their selection.

Internal validity measurement

With the aim of analyzing the internal validity of the different types of studies included in this review: PEDro internal validity scale was used with the clinical trials, STROBE internal validity scale was used with the

observational articles and the PRISMA scale was used with the reviews.

Study variables

The objective variables analyzed were: flexibility and range of motion (degree of stretching and/or range of motion of the injured lower limb compared with the contralateral leg and/or at the pre-injury time), functional tests (performance of sport-specific tests and functional measurement tests of the lower limb), strength (measurement of eccentric, concentric and/or isometric strength values of the affected lower limb, compared with the contralateral leg and/or pre-injury values), imaging tests (determination of the injury characteristics using diagnostic imaging tests), inflammation (presence or absence of signs of inflammation at the time of injury as a predictive value for recovery time) and injury history (previous history of injuries in the hamstring muscles).

At the same time, subjective variables were also analyzed, which aim to evaluate the athlete's perception towards their pathology from different points of view, such as pain (presence of pain at the time of the injury or in several hours after the injury, as well as when returning to activity), the ability to walk without pain (the time it takes the patient to walk without pain after the injury) and the athlete's subjective perception when carrying out the functional tests proposed.

Data synthesis and evidence hierarchy modeling

To maximize transparency, standardization, and rigor in our reporting, data extraction and synthesis were conducted following strict predefined parameters. Acknowledging the inherent heterogeneity of the included study designs—spanning randomized controlled trials, observational studies, and systematic reviews—we implemented a standardized evaluation approach to ensure consistent comparison across all methodological frameworks.

To establish a clinical decision-making framework, the data extracted from the selected studies were synthesized into a hierarchical model of importance, represented by a multidimensional overlapping diagram. The weighting of each variable—including Pain, Strength, Flexibility and Range of Movement, Functional Tests, and Imaging Tests—was determined by triangulating three fundamental factors. This methodological approach ensured that the resulting model is not merely a qualitative summary, but a weighted representation of return-to-play criteria, where the size and positioning of each domain reflect its empirical reliability and functional relevance within the current scientific literature:

Level of scientific evidence

Variables were prioritized based on their position within the evidence pyramid. Greater specific weight was assigned to criteria validated by systematic reviews (appraised via the PRISMA checklist) and randomized controlled trials (appraised via the PEDro scale), while findings from observational studies (appraised via the STROBE checklist) provided supplementary prognostic value.

Methodological internal validity

The individual scores of the studies, as detailed in the quality assessment tables, directly influenced the centrality of each variable. For instance, parameters frequently validated in high-scoring trials, such as biomechanical symmetry and functional capacity, were categorized as primary pillars.

Clinical consensus and relapse prediction

Data regarding the predictive capacity of each variable for re-injury risk—such as specific strength deficits or the persistence of imaging findings—were used to define the degree of overlap between clinical recovery and functional discharge.

Results

Figure 1 defines the process carried out to obtain the total number of articles. After performing the search using the aforementioned keywords, a total of 2482 articles were obtained, 2424 of which remained after discarding duplicates. Subsequently, 2350 were eliminated after reading the title and abstract to obtain a total of 74 articles analyzed. Finally, after subjecting them to the inclusion criteria that were previously specified, 45 articles were selected, eventually obtaining a final sample of 26 studies (13 of them are considered as a "main articles" and 13 of them as considered as a "secondary articles").

The analysis of the internal validity of the studies, presented in Table 1 using the PEDro scale, evidences a moderate methodological quality in the evaluated literature, with scores ranging between 3 and 6 out of a maximum of 10 evaluable points. The works by De Vos et al. (2014) and Jacobsen et al. (2016) stand out, both achieving the highest score in this group (6 points), sharing a solid foundation in their initial research design. Specifically, both trials show consistent compliance with items 2, 3, and 5, reflecting good control over fundamental methodological aspects such as randomization, allocation concealment, and blinding of all subjects, which provides highly positive initial reliability to their results.

On the other hand, Table 1 also highlights clear areas of opportunity to standardize and improve future trials in this line of research. A dispersion is observed in the compliance with relevant parameters during the execution

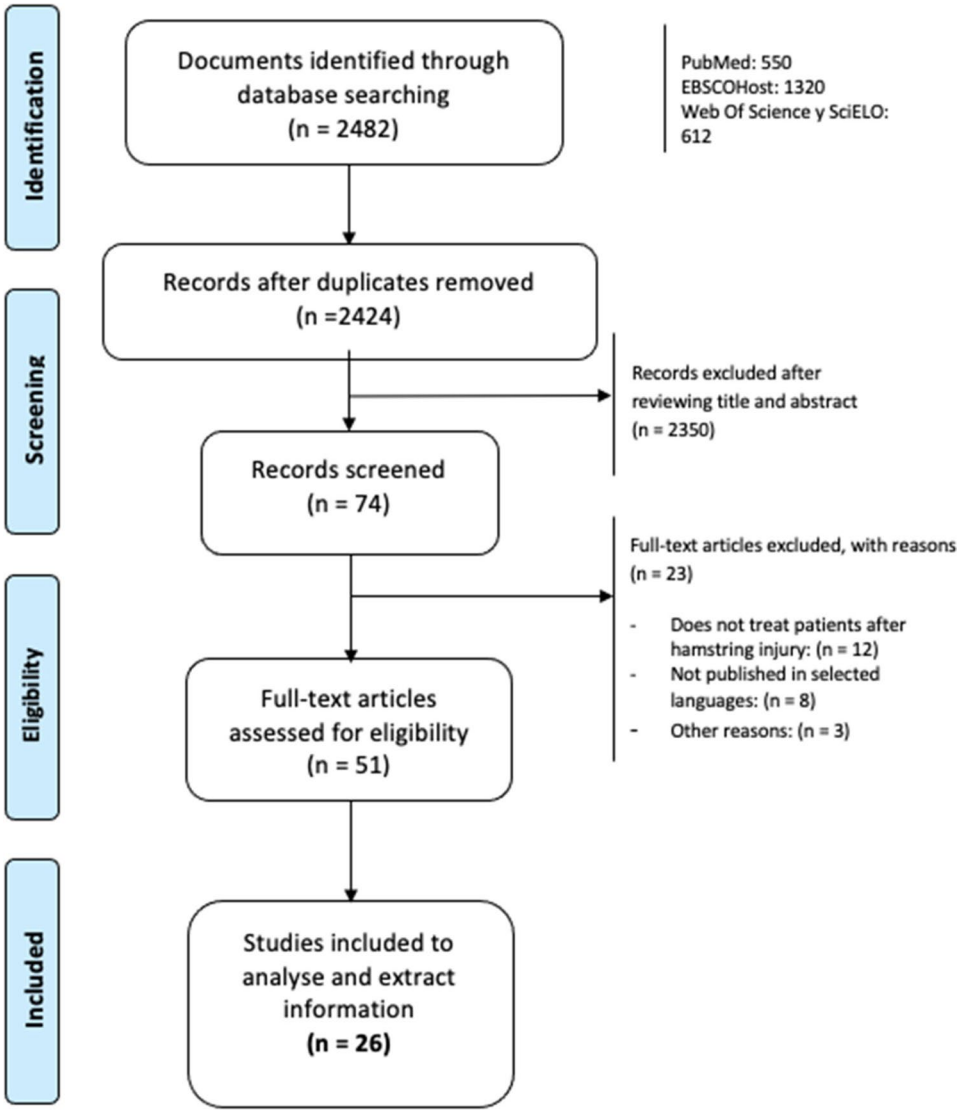


Fig. 1 Flowchart of the article selection process

Table 1 Internal validity analysis with PEDro scale

Articles	2	3	4	5	6	7	8	9	10	11	Total
De Vos R. et al. (2014) [36]	•	•	•	•	-	•	-	•	-	-	6
Jacobsen P. et al. (2016) [37]	•	•	-	•	•	-	-	-	•	•	6
Sanfilippo J. et al. (2013) [38]	-	-	-	-	-	-	•	•	-	•	3

and statistical analysis phases, such as the blinding of therapists and assessors (items 6 and 7), or the detailed presentation of between-group comparisons and point measures and measures of variability (items 10 and 11), aspects that are not addressed uniformly. Furthermore, guaranteeing adequate follow-up of the sample (item 8) is an element that, in this review, is only met in the study by Sanfilippo et al. [38]. Placing greater emphasis on unifying and detailing these methodological criteria in upcoming publications will help consolidate clinical rigor and greatly strengthen the applicability of the conclusions in this field.

The analysis of the internal validity of the selected studies, detailed in Table 2 using the STROBE checklist, shows a natural variability in the methodological reporting with an average score of 14.1 out of 22. The scores range from 8 to 19 points, evidencing that the level of detail in the reporting is diverse and independent of the articles' publication year. Across the board, there is great solidity in the structuring of the research. A high degree of compliance and consistency (items 2, 8, 12, 16, 17, and

Table 2 Internal validity analysis with STROBE checklist

Articles	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	Total
Askling et al. (2010) [39]	•	•	•	-	-	-	•	•	-	-	-	-	-	•	•	•	•	•	•	•	•	•	14
Ekstrand J. et al. (2016) [9]	•	•	•	-	-	•	-	•	-	•	•	•	•	-	•	-	-	•	•	-	-	•	13
Gonçalves-Valente H. et al. (2023) [40]	•	•	•	•	•	•	•	•	-	•	•	•	•	•	•	•	-	•	•	•	•	-	19
Guillodo et al. (2014) [41]	-	•	-	-	•	•	-	•	-	-	-	•	-	-	•	-	-	•	-	-	•	-	8
Gulgosteren E. et al. (2025) [42]	-	•	•	•	-	•	•	•	•	•	•	-	-	-	•	•	-	-	•	•	•	-	15
Jimenez-Rubio S. et al. (2019) [43]	•	•	•	•	-	-	-	•	-	-	•	•	-	-	-	•	•	•	-	•	•	-	12
Otsuka M. et al. (2022) [44]	•	•	•	•	-	•	•	•	•	-	•	•	-	•	•	•	•	•	•	-	•	•	18
Warren P. et al. (2010) [45]	-	•	-	-	•	•	•	•	-	-	•	•	-	•	•	•	•	•	•	•	•	•	16
Whiteley R. et al. (2022) [46]	-	•	-	-	•	•	-	•	-	-	•	•	-	-	•	•	•	•	•	-	-	-	11
Zambaldi M. et al. (2017) [47]	•	•	-	•	•	•	-	•	•	-	•	•	•	•	•	•	•	•	-	-	-	-	15

18) is noted in the formulation of the theoretical framework, the precise definition of the evaluation variables, and the transparent presentation of both statistical methods and main results, which adequately facilitates the reading and comprehension of the evaluated documents.

On the other hand, Table 2 also reflects certain areas of opportunity to standardize and enrich future publications in this field. A margin for improvement is identified in specific methodological details, such as the formal justification of the sample size (item 10), which is explicitly detailed in a minority of the works. Likewise, highly valuable aspects such as the reporting of bias control (item 9) or the discussion regarding the applicability and generalization of the results (item 21) show a lower frequency of compliance. A greater emphasis on these methodological elements in future research will contribute to consolidating the excellent descriptive base that this literature already possesses, increasing the robustness and overall rigor of the conclusions in this matter.

The analysis of the internal validity of the selected reviews, detailed in Table 3 using the PRISMA checklist, reflects a notable variability in the level of methodological reporting, with an average score of 5.5 out of a total of 27 evaluable items. A marked dispersion in the results is observed, ranging from 2 points to a highly outstanding case of 17 points achieved by the work of Van der Horst N. et al. (2016), which stands out significantly from the rest of the group due to its descriptive breadth. Across the board, the evaluated literature shows greater solidity in very specific sections; there is higher consistency of compliance in items linked to the introduction and closure of the documents, such as the formulation of a clear rationale (item 3) or the discussion of the study's limitations and conclusions (items 25 and 26), aspects that adequately facilitate the initial and final contextualization of the research.

On the other hand, Table 3 also highlights important areas of opportunity to standardize and enrich the structural quality of future publications in this field under the PRISMA guidelines. A wide margin for improvement is identified in the detailed description of the core methodological process, given that the vast majority of the intermediate items on the scale—related to aspects such as protocol registration, exact methods of data synthesis, or exhaustive detail in the presentation of individual results—show a minimal frequency of compliance in almost all the evaluated articles. Using the exhaustive level of detail exhibited by the highest-scoring study as a benchmark would represent an excellent strategy to unify these criteria in future works, greatly strengthening the transparency, rigor, and overall reproducibility of reviews in this subject area. There is no minimum recommended score that has been accepted in any of the three validity measurement scales used, but they consist of instruments

Table 3 Internal validity analysis with PRISMA checklist

PRISMA DOMAINS	Chu S. et al. (2016) [48]	Creighton D. et al. (2010) [27]	Erickson L. et al. (2017) [49]	Gismonti-García A. et al. (2022) [50]	Hickey T. et al. (2022) [51]	Macdonald B. et al. (2019) [52]	Martin *R.* et al. (2022) [53]	Mendiguchia J. et al. (2011) [54]	Perna *P.* et al. (2022) [55]	Perna *P.* et al. (2025) [56]	Pieters D. et al. (2021) [57]	Van der Horst *N.* et al. (2016) [25]	Ye J. et al. (2026) [58]
1	-	-	-	-	-	-	-	-	•	•	-	•	-
2	-	-	-	-	-	-	-	-	-	-	-	•	•
3	•	•	-	-	•	•	•	•	•	•	•	•	•
4	•	•	•	-	•	-	•	-	•	•	-	•	•
5	-	-	-	-	-	-	•	-	•	•	•	•	-
6	-	-	-	-	-	-	•	-	•	•	•	•	-
7	-	-	-	-	-	-	•	-	•	•	-	-	-
8	-	-	-	-	-	-	•	-	•	•	-	-	-
9	-	-	-	-	-	-	•	-	•	•	-	•	-
10	-	-	-	-	-	-	-	-	•	•	-	-	-
11	-	-	-	-	-	-	-	-	•	•	-	•	-
12	-	-	-	-	-	-	-	-	-	-	-	•	-
13	-	-	-	-	-	-	-	-	-	-	-	-	-
14	-	-	-	-	-	-	-	-	-	-	-	•	-
15	-	-	-	-	-	-	-	-	-	-	-	•	-
16	-	-	-	-	-	-	-	-	-	-	-	-	-
17	-	-	-	-	-	-	-	-	-	-	-	•	-
18	-	-	-	-	-	-	-	-	-	•	-	-	-
19	-	-	-	-	-	-	-	-	•	•	-	-	-
20	-	-	-	-	-	-	-	-	•	•	-	-	-
21	-	-	-	-	-	-	-	-	-	•	-	-	-
22	-	-	-	-	-	-	-	-	-	-	-	-	-
23	-	-	-	-	-	-	•	-	•	-	-	•	•
24	-	-	-	-	-	-	-	-	•	-	-	•	•
25	•	•	•	•	•	•	-	•	•	•	-	•	•
26	-	•	•	•	-	•	-	•	-	•	•	•	-
27	-	•	-	-	-	-	-	-	•	-	•	•	-
Total	3	5	5	2	3	3	8	3	16	16	6	17	6

that allow advice on the correct understanding by the reader of the types of articles analyzed.

Table 4 presents the identifying data of the primary studies included in this review, including the author and year, type of study, sample characteristics, sample size, intervention, outcome variables and results.

In addition, Table 5 shows the identifying data of the secondary studies, including the author and year, type of study, number of studies, variables analyzed, outcome measures and results.

In Fig. 2, it can be observed that the variables of pain, functional tests, strength, and flexibility and range of motion are the most frequently used as criteria for returning to activity after an injury to the hamstring muscles in the reviewed literature. On the other hand, the variables of inflammation and injury history are the least used and those with the lowest intensity. The larger size of a bubble refers to a greater presence of the variable in the analyzed studies, both primary and secondary.

The analysis of pain measurement reflects an extremely high clinical value in the reviewed literature, highlighting a strong and enriching consensus in placing this variable as the central axis for decision-making regarding the return to sporting activity. It is especially positive to observe how the evidence not only aligns around the standard of 'absence of pain', but also fosters a mature and highly practical debate; innovative approaches question the dogma of 'zero pain' by proposing more realistic and functional tolerance thresholds, while other authors manage to use initial pain as an excellent predictor for recovery prognosis, expanding the tools available to professionals. To further consolidate this magnificent clinical foundation, a methodological aspect that would present an interesting margin for improvement in future research would be a greater specification of the measurement instruments used, since standardizing the use of validated scales—currently detailed in only a minority of the works—would help homogenize reporting and greatly facilitate the exact comparison of these valuable clinical criteria (Table 6).

The analysis of flexibility and range of motion measurement reveals a great effort in the literature to clinically objectify physical recovery, highlighting a valuable functional consensus by establishing biomechanical symmetry—whether by reaching pre-injury values or matching the flexibility of the healthy contralateral leg, often with well-defined deficit margins, such as the 10% proposed by Van der Horst et al.—as the main standard for authorizing the return to activity. It is very positive to observe the wealth of specific tests analyzed, such as the Active Knee Extension Test or the rigorous Askling-H Test, which provides professionals with excellent quantitative tools to evaluate tissue functionality and associate specific deficits with prognosis, as wonderfully quantified by the work of De Vos et al. by calculating a 13% risk of relapse for each degree of deficit. To further polish and consolidate this solid framework, an area with an interesting margin for improvement for future publications would be to reduce heterogeneity and always specify the measurement instrument, since a block of studies still omits this methodological detail; furthermore, although setting 'pre-injury values' as a goal is ideal in theory, in clinical reality, these prior records of the athlete are often unavailable, so prioritizing and standardizing protocols based on direct comparison with the uninjured limb could prove to be a much more pragmatic and reproducible criterion for day-to-day practice (Table 7).

The analysis of muscle strength measurement presented in Table 8 reflects an appropriate level of clinical objectification in the literature, consolidating this variable as a relevant element for guiding the return to sporting activity. It is positive to observe the diversity of instrumental tools employed—such as dynamometers, the Nordbord system, or the single-leg bridge test—and the definition of precise prognostic thresholds, such as the repetition bands proposed by Erickson et al., or the calculation by De Vos et al. associating a 4% increase in relapse risk per Newton of force lost. Likewise, the consensus of using symmetry with the healthy contralateral limb as a reference standard, typically accepting a maximum deficit of 10%, provides a functional criterion of great practical utility. To further consolidate this methodological framework, future publications should systematically detail the instrument and the exact angulation of the test, as a subgroup of studies in Table 8 currently omits this information; given that force production is directly dependent on the joint angle and contraction type, specifying these technical aspects would improve the standardization and clinical reproducibility of the findings in daily professional practice.

The analysis of imaging measurement using Magnetic Resonance Imaging (MRI) presented in Table 9 highlights the significant diagnostic value of this tool while simultaneously fostering a relevant clinical debate regarding its predictive capacity for returning to activity. It is positive to observe how the evidence recognizes MRI as an essential instrument for characterizing "anatomical severity," allowing for the precise identification of edema size, the specific muscles involved, and the involvement of the tendon, which provides an objective baseline for tissue healing. However, the data in Table 9 also introduces a valuable nuanced perspective by showing that structural imaging findings do not always correlate directly with functional readiness; for instance, as noted by Sanfilippo et al. and Otsuka et al., significant signs of injury or high-intensity signals can persist even when an athlete is clinically ready to compete. To further refine this methodological area, future research could focus on

Table 4 Characteristics of included studies: main articles

Author and year	Type of study	Sample characteristics	*N*	Intervention	Outcome variables	Results
Askling et al. (2010)	Validity study	Healthy subjects [11] and elite athletes with acute hamstring injury [11]	22	-	Range of mobility	No statistically significant differences were observed. In the injured lower limb, active flexibility was significantly greater (23%) than passive flexibility. Hip flexion velocity was significantly greater in the uninjured (201 ± 29 °/s) than in the injured (175 ± 34 °/s). Insecurity (measured by the VAS scale) was 52 for the injured limb and 0 for the uninjured limb.
De Vos R. et al. (2014)	Controlled clinical trial	Patients between 18 and 50 years old, with a clinical and radiological diagnosis of acute injury to the hamstring muscles	64	Two 3 ml injections of platelet-rich plasma or normal saline at the injury site	Relapse rate	17 relapses (27%). Athletes with previous injuries to the hamstring muscles, a lack of strength in 15° knee flexion and the presence of pain on palpation in the hamstring muscles were related to the presence of relapse.
Ekstrand J. et al. (2016)	Prospective cohort study	First team players of UEFA teams	1488 injuries of the hamstring muscle	-	Examination procedure	917 (67%) were examined by imaging tests. Only grade 1 (173) and grade 2 (82) cases were included in the study.
					Reliability of assessments	There was perfect agreement between two independent radiologists in describing the grade; substantial agreement for the location and type of lesion; and substantial to almost perfect agreement for the measurement of edema.
					Degrees and days of absence	Absence time was significantly longer for grade 2 injuries than for grade 1 injuries.
					Musculature involved	212 (84%) affected the biceps femoris, 30 (12%) the semimembranosus muscle, and 11 (4%) the semitendinosus muscle. There was no significant difference in the absence time between the three different muscles.
					Lesion size and days of absence	In grade 1, there was a small to medium correlation between the time of absence and the extent of edema. In grade 2, there was a medium correlation with the absence time and the Z and X planes of the edema.
					Location of the injury and days of absence	In grade 1, the most common location was in the middle third (17 days of absence) and distal (19 days of absence). In grade 2, the most common location was in the middle third (23 days of absence).
					Type of injury and days of absence	No differences were found in days of absence between the different types of injury
Guillodo et al. (2014)	Cohort study	Athletes unable to continue their sports session due to pain in the back of the thigh	120	-	Time to return to activity	65 (54.2%) returned to pre-injury level in 40 days or less. 55 (45.8%) returned to pre-injury level in 40 days or more.
					Univariate analysis of return to activity criteria	Five criteria were associated with a late return to activity: pain intensity greater than 6 on the VAS scale, muscle tear sound, pain in daily life activities for more than 3 days, bruising, and limitation greater than 15° in range of motion compared to the contralateral leg.

Table 4 (continued)

Author and year	Type of study	Sample characteristics	*N*	Intervention	Outcome variables	Results
Jacobsen P. et al. (2016)	Controlled clinical trial	Female athletes with grade 1 and 2 hamstring injuries	90	Platelet-rich plasma injections and physical therapy. Five days/week until returning to activity	Time to return to activity	Variables associated with a longer return to activity: greater difference in isokinetic strength in the uninjured leg, increased pain at the time of injury, presence of pain when performing the single-leg bridge exercise, and delay in the start of physical therapy rehabilitation.
				Platelet-poor plasma injections and physical therapy. Five days/week until returning to activity		
				Physiotherapy. Five days/week until returning to activity		
Jimenez-Rubio S. et al. (2018)	Prospective longitudinal study	Professional footballers from the Spanish First Division	19	-	Aike´s V test	High validity rate for the rehabilitation program developed
Otsuka M. et al. (2022)	Cohort study	Sprinters who have suffered a hamstring injury	12	-	Time to return to activity	Range of 5 to 46 days to return to activity.
					Running biomechanics	The specific running tests had statistically significant differences. In the kinematics of running biomechanics, only half of the tests performed had statistically significant differences. In the kinetics of running biomechanics, all but one test (peak hip flexion moment) had statistically significant differences.
					Function of the hamstring muscles	The two tests performed (passive straight leg raise test and isometric knee flexion strength test) obtained statistically significant differences.
					Morphology of the hamstring muscles	Statistically significant differences were obtained in all the parameters analyzed.
Sanfilippo J. et al. (2013)	Controlled clinical trial	Recreational athletes with acute hamstring injury who have completed a rehabilitation program	25	Controlled rehabilitation program	Strength	At the time of return to activity, the injured leg shows a strength deficit compared to the counterlateral leg.
					Morphology	Upon return to activity, there is evidence of muscle injury.
Warren P. et al. (2010)	Prospective observational study	Elite Australian football players who have suffered a hamstring injury		-	Time to return to activity	Range from 1 to 8 weeks, with an average of 26 days.
					Relapse	9 (15%) relapsed within 3 weeks. 6 of them (66%) relapsed within the first week.

Table 4 (continued)

Author and year	Type of study	Sample characteristics	*N*	Intervention	Outcome variables	Results
Whiteley R. et al. (2022)	Observational study	Professional footballers in the Qatar League have their first hamstring injury of the season	38	-	Physical performance in the match through a multi-camera system	Decrease in maximum sprint speed by −0.25 km/h Decrease in distance traveled at maximum speed of −43 m. Decrease in sprint distance from − 22 m.
Zambaldi M. et al. (2017)	Consensus study	Physicists and physiotherapists working in the English Premier League	20	-	Criteria for returning to activity	12 criteria achieved consensus with criteria for return to activity. They achieved 100% consensus: the ability to develop a sprint at maximum speed, the ability to reach maximum linear speed, the development of a progressive running training plan, completion of at least one football-specific test, subjective perception of safety upon return to activity, complete recovery of strength, and complete recovery of flexibility.

integrating these objective imaging findings with functional biomarkers, as this would help clarify the discrepancy between anatomical recovery and clinical discharge, ultimately leading to more personalized and safer return-to-sport protocols.

The analysis of functional measurements using running-specific tests and Aiken's V, as presented in Table 10, underscores the critical importance of transitioning from isolated clinical markers to dynamic, sport-specific performance criteria. It is highly positive to observe a strong consensus across the literature regarding the necessity of high-intensity functional testing—such as maximum speed sprints, acceleration-deceleration patterns, and sport-specific movements—to ensure a safe return to activity. The studies in Table 10 highlight that successful rehabilitation is not merely defined by the absence of symptoms, but by the restoration of pre-injury functional capacity, with several authors specifically recommending the use of batteries that simulate match conditions and positional demands. To further enhance this comprehensive approach, future research could benefit from more precise standardization of these "functional batteries," as a significant portion of the articles grouped under the "not specified" category in Table 10 describe the concepts of symmetry and sport-specific exercises without always defining the exact protocols or passing scores. Refining these details would help clinicians bridge the gap between general rehabilitation and the highly specialized demands of competitive play, ensuring that functional recovery is both measurable and reproducible across different sporting contexts.

The analysis of exercise progression, as detailed in Table 11, highlights a fundamental clinical focus centered on the gradual increase of mechanical stimulus as a key to successful recovery. It is very positive to observe the existing consensus regarding the importance of a structured transition, which evolves from low-load exercises to high-intensity tasks and sport-specific movements. The data in Table 11 shows that the evidence prioritizes load control and respect for biological healing times, utilizing progression criteria that allow the athlete to adapt progressively to the demands of competition.

To further complement this methodological foundation, an aspect that could be detailed with greater depth in future research is the definition of objective indicators required to move from one exercise phase to the next. While Table 11 reflects a clear structure regarding the typology of exercises, the specification of quantitative thresholds—such as specific strength levels, running volumes, or GPS metrics—to validate each phase transition would help reduce subjectivity in decision-making. Refining these transition criteria would enable professionals to design even more precise and safe rehabilitation programs, minimizing the risk of relapse during the most critical stages of functional progression.

Figure 3 shows that the criterion of using functional tests to determine the return to activity after a hamstring injury is fully supported by all the included articles that have analyzed this variable as a criterion for returning to sports (eight articles) [9, 47, 49, 53, 58, 59, 61]. The larger size of a bubble refers to a greater presence of the criterion for return to activity in the analyzed studies, both primary and secondary. However, the second criterion with the largest number of articles supporting it is the absence of pain, but not all the analyzed works are in favor of its use as a criterion for returning to activity, with two of them [47, 51] being against its inclusion (ten articles in favor, 83%) [25, 27, 36, 39, 40, 52, 55, 56, 59, 61]. Thus, the criteria for returning to activity after a hamstring injury supported by all the analyzed articles was similar flexibility with respected to pre-injury levels (seven articles) [25, 27, 39, 44, 47, 52, 56, 61], similar

Table 5 Characteristics of included studies: secondary articles

AUTHOR AND YEAR	TYPE OF STUDY	VARIABLE ANALYZED	OUTCOME MEASURES	RESULTS
Chu S. et al (2016)	Bibliographic review	Diagnostic	History	Important for proper diagnosis
			Test	
			Imaging	It is not related to the return to activity forecast
			Differential diagnosis	Hamstring avulsion injury, ischial process avulsion, stress fractures, adductor muscle strain, referred pain from the lumbar spine, hip, or sacroiliac joint, and ischial gluteal bursitis.
		Treatment	Acute Hamstring Strain	Includes eccentric and progressive range of motion training
			Proximal Hamstring Tendinopathy	Eccentric strength training
		Return to activity	Acute Hamstring Strain	The type of muscle injury and clinical factors
			Proximal Hamstring Tendinopathy	Full range of motion and full recovery of strength
Creighton D. et al (2010)	Bibliographic review	Return to play	Step 1: Evaluation of Health Status	Demographic information, symptoms, personal medical history, signs, laboratory tests, functional tests, psychological status and potential harm.
			Step 2: Evaluation of Participation Risk	Type of sport, playing position, dominant leg, competitive level and protection ability
			Step 3: Decision Modification	Timing of the season, pressure from the athlete, external pressure, masking the injury, conflicts of interest and fear of litigation
Erickson L. et al (2017)	Bibliographic review			Hamstring injuries are common in athletes and need proper rehab to prevent re-injury and restore performance. Effective programs include neuromuscular control, agility, core stability, and eccentric training. Even after returning to sport, continued rehab is important to reduce re-injury risk.
Gismonti-García A. et al (2022)	Bibliographic review			Hamstring injuries are complex and require individualized evaluation in football players to identify risk and apply effective, evidence-based prevention and rehab programs. Early diagnosis and goal-based rehab are key, though no clear consensus exists on diagnosis methods. Ongoing monitoring, load management, and prevention strategies are essential to reduce both initial and recurrent injuries.
Hickey T. et al (2022)	Bibliographic review			Dr. David A. Opar is a co-inventor of the NordBord, a device for testing eccentric hamstring strength, and has received revenue from its commercialization. He is a minority shareholder in Vald Performance, the company that markets the device, and has received unrelated research funding from them. He chairs their research committee (unpaid) and has family members involved with the company as shareholders or employees.
Macdonald B. et al (2010)	Bibliographic review			This paper presents British Athletics' structured approach to hamstring injury rehab, based on the BAMIC (British Athletics Muscle Injury Classification) framework. Their method has reduced re-injury rates, and they suggest that sport-specific rehab strategies are more effective than a one-size-fits-all approach
Martin R. et al (2022)	Bibliographic review			Recovery from a hamstring injury involves reducing pain early on, then building strength, flexibility, and balance, followed by sport-specific training. A full return to sport requires being pain-free, having full strength and movement, and completing sport-specific tasks. Recovery may take weeks to months, and exercises like the Nordic hamstring can help prevent future injuries.
Mendiguchia J. et al (2010)	Bibliographic review			Return-to-sport rehab based only on subjective measures like "pain-free movement" may leave hidden deficits that increase re-injury risk. A criteria-based approach using objective tests can better detect and address these issues through a structured progression. While promising, more research is needed to validate this method.
Perna P. et al (2022)	Scoping review	Return to higher speeds	Absence of pain	Before progressing to rehabilitation on the field and more intense specific activities, one must be able to complete functional activities, establish bonds and progress through a structured rehabilitation process in the absence of pain.
			Hamstring strength	When establishing the criteria for return to activity, there must be a difference of less than 10% in the force exerted between the injured and uninjured limb.
			Imaging	There are no specific guidelines or measures in place when using imaging tests as a return-to-activity criterion.

Table 5 (continued)

AUTHOR AND YEAR	TYPE OF STUDY	VARIABLE ANALYZED	OUTCOME MEASURES	RESULTS
Perna P. et al (2025)	Scoping review	Return to practice criteria	Absence of pain	It is the most commonly used criterion for resuming activity and consists of the absence of pain upon palpation.
			Hamstring flexibility	Active and passive measurement of flexibility is an important criterion for returning to activity.
			Strength	Highlight the importance of eccentric strength as a criterion for return to activity.
Pieters D. et al (2021)	Bibliographic review			Current practices often overlook biological healing time when deciding return-to-play (RTP) after a hamstring injury. Research shows athletes often return before full tissue healing, increasing re-injury risk. A more effective approach includes a comprehensive RTP test battery and respect for natural healing. Clinicians should ensure the tissue is biologically ready, not just functionally. A minimum of 4 weeks is recommended before RTP for grade 1 and 2 hamstring injuries.
Van der Horst N. et al (2016)	Qualitative systematic review	Return to activity	Definition of return to activity	Half of the included articles provided a definition of return to play
			Definition of return to activity	None of the return to activity criteria have been validated
Ye J. et al (2026)	Bibliographic review	Return-to-Sport	Definition of return to sport	The protocol for returning to physical activity includes objective functional measures, such as pain resolution, normal range of motion, and strength symmetry.

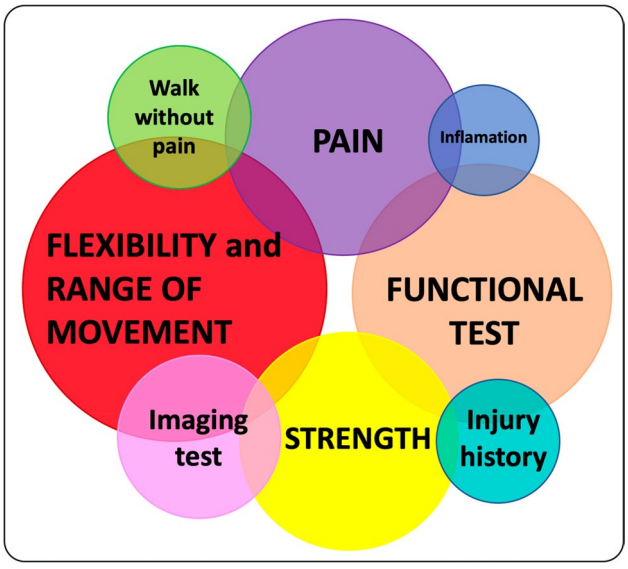


Fig. 2 Importance of the different criteria used for returning to activity according to the number of repetitions

strength with respect to pre-injury levels (seven articles) [25, 27, 42, 44, 47, 56, 61], injury history (four articles) [25, 36, 45, 53], maximum speed tests similar to pre-injury values (three articles) [44, 46, 52], similar flexibility with the contralateral leg (three articles) [27, 49, 56] and time to walk without pain after the injury (2 articles) [45, 48]. On the other hand, the criterion for returning to activity with the lowest percentage of articles in favor (33%) is that of imaging tests, since there are more articles against its implementation as a criterion for returning to activity (six articles) [37, 38, 44, 48, 55, 60] than those supporting it (three articles) [9, 27, 40].

Discussion

The present systematic review aimed to analyze the tests to determine the return to activity after a hamstring injury. This wide range of variables described requires that a wide variety of articles be analyzed, in order to reach all the scientific literature published to date and thus be able to obtain conclusive results in this review. In the existing literature, it was observed that the variables of pain, flexibility and range of motion, functional tests and strength were the most studied and the most frequently used as criteria for returning to activity. At the same time, the variability in the existing tests and in the parameters applied to them hinders analyses and comparisons in this field of study.

General interpretation of results

The primary purpose of this systematic review was to establish defined criteria for clinical decision-making regarding the return to play (RTP) following a hamstring muscle injury. The findings suggest a paradigm shift: functional discharge has moved away from being a process based exclusively on biological timelines toward a multifactorial, hierarchical, and performance-based model, where biomechanical symmetry and dynamic functional capacity take precedence over isolated anatomical resolution. However, the interpretation of these findings must be contextualized within the significant heterogeneity of study types and the inconsistent methodological quality observed across the literature. By

Table 6 Pain measurement using Numeric Pain Scale and Visual Analog Scale

MEASUREMENT INSTRUMENT	ARTICLE	OBTAINED RESULTS	CRITERION
Numeric Pain Rating Scale (0–10)	Hickey J. et al. (2022) [51]	Pain score ≤ 4 during exercise at the time of returning to activity	Pain reduction as a criterion for returning to activity.
	Gonçalves-Valente H. et al. (2023) [40]	The absence of pain upon palpation, contraction, and stretching is the most commonly used criterion for return to activity. The numerical rating scale is the most frequently used measurement tool.	Absence of pain as a criterion for returning to activity.
Visual Analog Scale (VAS)	Guillodo Y. et al. (2014) [41]	Score higher than 6 on initial assessment is associated with longer recovery time	Relationship between initial pain and time to return to activity.
	Askling C. et al. (2010) [39]	A score of 0 (no feeling of insecurity) when performing the flexibility tests upon returning to activity reduces the possibility of relapse to 0%.	Absence of pain as a criterion for returning to activity.
Not specified	Gismonti-García A. et al. (2022) [50]	Absence of pain upon return to activity	Absence of pain as a criterion for returning to activity.
	Erickson L. et al. (2017) [49]	Absence of pain upon return to activity	Absence of pain as a criterion for returning to activity.
	Jacobsen P. et al. (2016) [37]	The greater the pain at the time of the injury, the longer the recovery time will be	Relationship between initial pain and time to return to activity.
	Zambaldi M. et al. (2017) [47]	83.33% of participants reported no pain as a criterion for returning to activity. However, the author disagrees on the concept of "no pain" in returning to activity.	Pain reduction as a criterion for returning to activity.
	Creighton D. et al. (2010) [27]	Absence of pain upon return to activity	Absence of pain as a criterion for returning to activity.
	Pieters D. et al. (2021) [57]	Absence of reproducible pain on palpation and contraction upon returning to activity.	Absence of pain as a criterion for returning to activity.
	Van der Horst N. et al. (2016) [25]	Absence of pain on palpation and during functional tests as the most commonly used criterion to determine return to activity.	Absence of pain as a criterion for returning to activity.
	Hickey J. et al. (2017) [59]	Absence of pain on palpation of the painful area and absence of pain when performing sprints as a criterion for returning to activity.	Absence of pain as a criterion for returning to activity.
	Martin R. et al. (2022) [53]	Absence of pain during sport-specific movements and absence of pain in the hamstring muscles after running at maximum speed.	Absence of pain as a criterion for returning to activity.
	Perna P. et al. (2022) [55]	The absence of pain was recognized as the main criterion for return to activity for the majority of participants (57%), both as a clinical assessment measure and in physical tests.	Absence of pain as a criterion for returning to activity.
	Perna P. et al. (2025) [56]	The absence of pain was indicated as the most important criterion for determining the return to activity, both in the palpation of the hamstring muscles, in strength and flexibility tests, and in sport-specific functional tests.	Absence of pain as a criterion for returning to activity.

applying rigorous transparency and standardized reporting in our hierarchical model, we were able to filter and weight the criteria appropriately, ensuring that the most clinically reliable variables stand out despite the varying quality of the primary sources.

Our evidence synthesis is articulated through two complementary visual models. On one hand, Fig. 2 positions functional tests and flexibility/range of motion as the central pillars with the greatest specific weight. This hierarchy is based on the high internal validity of the analyzed studies (Tables 1 and 2, and 3); variables validated by high-quality systematic reviews (such as Van der Horst et al. in Table 3) and clinical trials with high scores on the PEDro scale (such as De Vos et al. and Jacobsen et al. in Table 1) possess greater centrality than the findings of observational studies. On the other hand, Fig. 3 represents the magnitude of these criteria according to their frequency of repetition in the literature. In this model, the variable "Walking without pain" stands out prominently as the most cited criterion, followed closely by "Functional Tests." This recurrence underscores that, for the scientific community, basic pain-free functionality and the execution of specific sporting movements are the indispensable minimum requirements for initiating any advanced phase of RTP.

Regarding symmetry parameters, data from Tables 7 and 8 consolidate the need to restore physical capacity relative to the contralateral limb or pre-injury levels. Consensus points toward a maximum deficit threshold of 10% in flexibility and strength. The relevance of this data is critical: according to the calculations of De Vos et al. (Table 8), each Newton of force lost is associated with a 4% increase in the risk of relapse. However, Fig. 3 also introduces variables such as "Previous injury history," whose repeated mention in the literature alerts clinicians to the need for a more conservative approach to exercise

Table 7 Flexibility and range of motion measurement

MEASUREMENT INSTRUMENT	ARTICLE	OBTAINED RESULTS	CRITERION
Passive Straight Leg Raise Test	Otsuka M. et al. (2022) [44]	There is an association between the degree of flexibility of the hamstring muscles and the time of return to activity. Dysfunction in the flexibility of the hamstring muscles is observed with respect to pre-injury values.	Equality to pre-injury flexibility values.
	Macdonald B. et al. (2019) [52]	Obtaining values equal to those before the lesion	Equality to pre-injury flexibility values.
	Warren P. et al. (2010) [45]	No relationship is found between the values obtained in this test and the presence of relapse in hamstring muscle injuries.	There are no established flexibility criteria.
	Chu S. et al. (2016) [48]	Deficits in the "Passive Straight Leg Raise Test" values are significantly associated with the time to return to activity after a hamstring injury.	Recommendation of a specific test as a criterion for returning to activity.
	Zambaldi M. et al. (2017) [47]	Complete recovery of hamstring flexibility is necessary for a safe return to activity.	Equality to pre-injury flexibility values.
Active Knee Extension Test	Zambaldi M. et al. (2017) [47]	Complete recovery of hamstring flexibility is necessary for a safe return to activity.	Equality to pre-injury flexibility values.
	Mendiguchia J. et al. (2011) [54]	Recommends the active knee extension test to measure hamstring flexibility and eliminate neural tension.	Recommendation of a specific test as a criterion for returning to activity.
	De Vos R. et al. (2014) [36]	The larger the number of degrees of deficit in the "Active Knee Extension Test", the greater the risk of relapse (13% risk for each degree of deficit).	Equality to pre-injury flexibility values.
	Warren P. et al. (2010) [45]	Difference greater than 10° was close to reaching a statistically significant difference.	Equality to pre-injury flexibility values.
Active Straight Leg Raise Test	Zambaldi M. et al. (2017) [47]	Complete recovery of hamstring flexibility is necessary for a safe return to activity.	Equality to pre-injury flexibility values.
Askling-H Test	Macdonald B. et al. (2019) [52]	Negative Askling-H Test	Equality to pre-injury flexibility values.
	Hickey J. et al. (2017) [59]	The Askling-H Test has been shown to be reliable and sensitive in measuring active range of motion. Combined with traditional return-to-activity criteria, it is associated with a decreased risk of relapse.	Recommendation of a specific test as a criterion for returning to activity.
Active Hamstring Flexibility Test	Askling C. et al. (2010) [39]	Active flexibility similar to the uninjured leg was observed, but not passively. The new ballistic hamstring flexibility test showed greater ability to detect signs of relapse before returning to activity.	Same flexibility values as those of the contralateral leg. Recommendation of a specific test as a criterion for returning to activity.
Not specified	García A. et al. (2022) [50]	Hamstring flexibility similar to pre-injury values of the contralateral leg (10 ~ 15%)	Equality to pre-injury flexibility values. Same flexibility values as those of the contralateral leg.
	Guillodo Y. et al. (2014) [41]	Limitation in range of motion greater than 15° with respect to the contralateral leg is associated with longer recovery time	Same flexibility values as those of the contralateral leg.
	Creighton D. et al. (2010) [27]	Range of motion equal to or close to pre-injury values or symmetrical with the unaffected limb.	Same flexibility values as those of the contralateral leg. Equality to pre-injury flexibility values.
	Van der Horst N. et al. (2016) [25]	The criterion for returning to activity is established as normal hamstring flexibility, defined as a deficit of less than 10% between the affected leg and the unaffected leg.	Same flexibility values as those of the contralateral leg.
	Perna P. et al. (2025) [56]	Flexibility should be measured using active and passive tests. A difference of 0%–10% between the injured and uninjured leg, or compared to pre-injury values, constitutes a criterion for return to activity.	Same flexibility values as those of the contralateral leg. Equality to pre-injury flexibility values.

progression (Table 11) when the athlete has a history of injury, due to alterations in muscle architecture.

A central point of debate is the role of imaging tests. As observed in Fig. 3, the "Imaging test" occupies a peripheral and smaller position. Data extracted from Table 9 (Otsuka et al., Sanfilippo et al.) demonstrate a "diagnostic paradox": the persistence of edema or high-intensity signals on MRI does not necessarily correlate with functional incapacity. Therefore, while initial diagnosis is key for staging anatomical severity, its use as a discharge criterion is questionable unless accompanied by a satisfactory functional battery (Table 10) that includes maximum speed sprints and changes of direction.

Finally, the integration of all information suggests that RTP success lies in a structured load progression (Table 11) that validates both tissue recovery and neuromuscular competence. Consistency between the analyzed methodological quality and the frequency of reported

Table 8 Strength measurement

MEASUREMENT INSTRUMENT	ARTICLE	OBTAINED RESULTS	CRITERION
Isometric Knee Flexion Strength Test	Otsuka M. et al. (2022) [44]	There is a relationship between the existing alteration in the strength of the hamstring muscles and the time of return to activity after an injury to said muscles.	Relationship between initial strength and time of return to activity.
Dynamometer	Jacobsen P. et al. (2016) [37]	A large difference in knee flexor strength at 60° of flexion compared to the uninjured leg is associated with a longer return to competition. Changes in strength developed in the mid-range test at 7 days are a predictor of return to activity.	Relationship between initial strength and time of return to activity.
	Zambaldi M. et al. (2017) [47]	Full recovery of strength compared to the uninjured leg and/or pre-injury values is necessary for a safe return to competition.	Same strength values as those of the contralateral leg. Equality in pre-injury strength values.
	Sanfilippo J. et al. (2013) [38]	A hamstring muscle strength deficit compared to the uninjured leg was observed in 10% of participants at the time of return to activity after a muscle injury. This strength deficit was related to the remaining muscle injury observed on MRI.	Relationship between initial strength and time of return to activity.
	Hickey J. et al. (2017) [59]	Studies that implement the use of the dynamometer show a decrease in the time of return to activity compared to those that use a combination of clinical and functional measurements.	Recommendation of a specific test as a criterion for returning to activity.
Single Leg Bridge	MacDonald B. et al. (2019) [52]	Capacity greater than 30 when performing the one-legged bridge	Recommendation of a specific test as a criterion for returning to activity.
	Erickson L. et al. (2017) [49]	It includes performing the single-leg hamstring bridge as a criterion for returning to activity, although it specifies that it increases the risk of suffering an injury to this muscle. It establishes that a score of less than 20 repetitions is considered poor, 25 repetitions is considered average, and more than 30 repetitions is considered good.	Recommendation of a specific test as a criterion for returning to activity.
Nordbord	Macdonald B. et al. (2019) [52]	Performance at a pre-injury level	Equality in pre-injury strength values.
Isometric Knee Flexion Resistance in 15º	De Vos R. et al. (2014) [36]	Athletes with a strength deficit in isometric knee flexion at 15º are more likely to suffer a relapse. There is a 4% increase per loss in Newtons.	Equality in pre-injury strength values.
Surface electromyography	Gulgosteren E. et al. (2025) [42]	Athletes who had suffered a hamstring injury showed less activation in that muscle compared to athletes without any injury, as well as between the injured and uninjured legs. This decreased activation was primarily evident during the eccentric phase of the contraction. This resulted in a 33% increase in the relapse rate.	Equality in pre-injury strength values.
Not specified	Creighton D. et al. (2010) [27]	Strength equal or similar to pre-injury levels, or symmetrical with respect to the unaffected leg.	Equality in pre-injury strength values. Same strength values as those of the contralateral leg.
	Hickey J. et al. (2022) [51]	Development of similar strength in the affected leg compared to the unaffected leg, considering a 10% deficit as similar.	Same strength values as those of the contralateral leg.
	Van der Horst N. et al. (2016) [25]	Development of similar strength in the affected leg compared to the unaffected leg, considering a 10% deficit as similar.	Same strength values as those of the contralateral leg.
	Perna P. et al. (2022) [55]	It establishes a difference in force exerted between the injured and uninjured leg of less than 10% as a criterion for return to activity. There is no consensus on the importance of eccentric strength for return to activity.	Same strength values as those of the contralateral leg.
	Perna P. et al. (2025) [56]	Although the importance of eccentric strength training as a protective mechanism against hamstring injuries has been demonstrated, there is no consensus on determining the measurement of eccentric muscle strength as a criterion for return to activity. A 10% difference in strength exerted is generally considered a return-to-activity criterion.	Equality in pre-injury strength values.

criteria allows for the conclusion that a safe discharge must prioritize the restoration of symmetrical strength, the absence of pain during walking, and, above all, the successful completion of functional tests that replicate the actual intensity of competition. This study provides an objective framework that allows clinicians to prioritize variables with the greatest scientific impact, optimizing athlete safety and reducing uncertainty in one of the

Table 9 Imaging measurement using MRI (Magnetic Resonance Imaging)

MEASUREMENT INSTRUMENT	ARTICLE	OBTAINED RESULTS	CRITERION
MRI (Magnetic Resonance Imaging)	Otsuka M. et al. (2022) [44]	No relationship was found between MRI imaging characteristics and return to activity, showing that high intensity signals were not completely eliminated.	Absence of relationship between imaging tests and return to activity.
	Reurink G. et al. (2015) [60]	Established "anatomical severity," measured by MRI, as one of the determining factors in returning to activity after a hamstring injury.	Establishing imaging tests as a criterion for returning to activity
	Ekstrand J. et al. (2016) [9]	The use of MRI allows distinguishing the different degrees, the size of the edema and the specific muscle of the hamstring affected (with the biceps femoris being the one with the highest rate of relapse), which have an impact on predicting the return to activity. They also allow observing the location of the injury and the type of muscle structure affected, although the first of these has no predictive value for the return to activity and the second is in question.	Establishing imaging tests as a criterion for returning to activity
	Macdonald B. et al. (2019) [52]	In the case of a muscle injury involving the tendon, MRI allows observing the structural recovery of said structure, providing additional information in the decision-making process regarding returning to activity.	Establishing imaging tests as a criterion for returning to activity
	Sanfilippo J. et al. (2013) [38]	At the time of return to activity, the affected muscle area still showed signs of injury in 20% (on average) of the affected area.	Establishing imaging tests as a criterion for returning to activity
	Gonçalves-Valente H. et al. (2023) [40]	Having an MRI scan right after suffering the injury is a very important prognostic factor in determining the return to activity.	Establishing imaging tests as a criterion for returning to activity
	Perna P. et al. (2022) [55]	Although MRI has been established as the standard criterion for the diagnosis of muscle injuries, there is no consensus to establish a relationship between MRI and return to activity.	Absence of relationship between imaging tests and return to activity.
Not specified	Creighton D. et al. (2010) [27]	Imaging techniques, including X-rays, MRI, CT, bone scans, and ultrasound, provide objective evidence of tissue healing and allow us to detect structural and physiological abnormalities that may indicate incomplete recovery.	Establishing imaging tests as a criterion for returning to activity

injuries with the highest recurrence rates in professional sports.

Pain

The absence of pain in palpation has been established as a criterion to be taken into account for progression in the recovery process and to determine the return to activity [56, 59], having been defined as the most referenced criterion for return to activity [40, 55, 62]. In addition, it has demonstrated a sensitivity of 90% in the diagnosis of injury in the hamstring muscles [63]. Moreover, it has been determined that the presence of pain on palpation at the time of return to activity increases the risk of relapse by almost four times, compared to the absence of pain [36], thus it could be compatible with an incomplete recovery process [64].

On the other hand, the presence of pain on contraction, especially in resisted knee flexion with the hip and knee flexed, has been established as a criterion for returning to activity and related to the time of absence [63, 65]. Specifically, it was observed that there was an increase of 4.7 days between those athletes with pain on contraction and those who did not present it [65].

There are divergences regarding the definition of absence of pain, since, on the one hand, it has been determined that a pain less than or equal to 4 on the Visual Analogue Scale (VAS) is considered safe [51], favoring early exposure and increasing the stimulus to contraction [66]. On the other hand, other authors determine, as a criterion for returning to activity, the absence of pain upon contraction [67] and at maximum stretch [68].

These differences in the conception of the need for absence of pain as a criterion for returning to activity could be determined by the presence of pain in the athlete´s daily life [69]. A prevalence of hip pain of 97.2% was determined in hockey players and 81.9% of knee pain in soccer players [70]. The existence of good pain and bad pain has also been determined, with the latter being more frequent with the presence of fatigue and presenting worse adaptive responses in the athlete [71]. At the same time, and in relation to what was mentioned above, the largest number of muscle injuries occur when the muscle is fatigued [72–74].

It is worth highlighting the importance of the pain variable in predicting the time of return to activity, since the greater the intensity of the pain, the longer the time of absence [63, 75]. In this sense, Wangensteen et al. [65] specified that each point added to the VAS scale at the time of injury increases the time of return to activity by 1.6 days. Guilodo et al. [66] determined that a score higher than 6 on the VAS for more than 3 days requires more than 40 days to return to competitive activity. In

Table 10 Functional measurement using Running-specific test and Aiken´s V

MEASUREMENT INSTRUMENT	ARTICLE	OBTAINED RESULTS	CRITERION
Running-specific test	Otsuka M. et al. (2022) [44]	Completion of two 60-metre runnings at maximum speed. The correct completion of this running test is determined to be one of the main criteria for returning to activity.	Recommendation of a specific test as a criterion for returning to activity.
	Whiteley R. et al. (2022) [46]	One of the main criteria for returning to activity after a hamstring injury is to restore the ability to run at maximum speed prior to the injury.	Equality in pre-injury functional values.
Aiken´s V	Jimenez-Rubio S. et al. (2019) [43]	The "Functional Sports Re-education" criterion was established, which constitutes one of the main criteria for returning to activity and the main objective of the rehabilitation program. The highest scores were obtained in those that include repeated sprint skills, re-education and retraining of acceleration-deceleration patterns, and those that involve movements specific to football and the position of the players (the study population were footballers).	Recommendation of a specific test as a criterion for returning to activity.
Not specified	Gismonti-García A. et al. (2022) [50]	It establishes as a criterion for returning to activity the development of functional tests, including those specific to the position, GPS (global positioning system) and specific match rehabilitation.	Recommendation of a specific test as a criterion for returning to activity.
	Erickson L. et al. (2017) [49]	Functional tests that incorporate sport-specific movements, with intensity and speed close to maximum, and without pain, limitation or instability should be included as criteria for returning to activity.	Recommendation of a specific test as a criterion for returning to activity.
	Macdonald B. et al. (2019) [52]	It establishes the correct progression in running and high-intensity training as a criterion for returning to activity.	Recommendation of a specific test as a criterion for returning to activity.
	Creighton D. et al. (2010) [27]	It includes, as criteria for returning to activity, the performance of functional tests, where we can find the performance of exercises that simulate the specific movements of the sport and an activity close to symmetry compared to the contralateral limb.	Recommendation of a specific test as a criterion for returning to activity.
	Van der Horst N. et al. (2016) [25]	A battery of functional tests is established as criteria for returning to activity after a hamstring injury, including: - Better sprint times compared to the pre-injury period. - Development of training and activity without restrictions. - Sport-specific movements at a level close to maximum intensity and speed. - Pain-free triple jumps with one leg and the Illinois agility test in 18.4 s.	Recommendation of a specific test as a criterion for returning to activity. Equality in pre-injury functional values.
	Martin R. et al. (2022) [53]	The criterion for returning to activity is established as the requirement to complete sport-specific movements at maximum speed and without pain in the hamstring muscles.	Recommendation of a specific test as a criterion for returning to activity.
	Pieters D. et al. (2021) [57]	The battery of return to activity criteria variables includes the performance of functional tests and the ability to perform sport-specific movements.	Recommendation of a specific test as a criterion for returning to activity.
	Hickey J. et al. (2017) [59]	It includes functional tests, such as the ability to sprint and perform sport-specific movements, although it states that it does not provide additional information on the risk of relapse.	Recommendation of a specific test as a criterion for returning to activity.
	Ye J. et al. (2026) [58]	The presence of deficits in parameters related to functional capacity determines the need to incorporate the measurement of sport-specific patterns and the performance of neuromuscular control exercises as return-to-activity criteria	Equality in pre-injury functional values.

this regard, although the absence of pain does not imply the absence of tissue involvement in imaging tests, the opposite case has been demonstrated in that the presence of pain has a strong relationship with the visualization of tissue damage, an aspect that is associated with an increase in the period of sick leave due to injury.

Flexibility and range of motion

A total of 13 articles included hamstring flexibility and range of motion as criteria to determine the return to activity. Some of these articles determined that failure to fully recover this parameter negatively influences the risk of a possible relapse [25, 27, 33, 36, 41, 44, 45, 47, 48, 52, 54, 59, 61]. In relation to this variable, two approaches can be found to determine full recovery. On the one hand, there are different studies in which the achievement of flexibility values similar to those before the injury is determined [36, 44, 45, 47, 52] and, on the other hand, obtaining values similar to those of the contralateral leg [25, 39, 41]. Despite these two well-defined approaches, there is no consensus on which of them to apply when returning to activity, although there are studies that established both criteria to determine complete recovery [27, 56, 61]. When this difference in flexibility between both legs is greater than 10%, it has been considered a risk factor for muscle injury [76].

Table 11 Other measurements: injury history, pain-free walking and inflammation

MEASUREMENT INSTRUMENT	ARTICLE	OBTAINED RESULTS	CRITERION
Injury history	Warren P. et al. (2010) [45]	Injury history is a significant independent predictor of return to competition time. It also predisposes to future injuries in said muscles and serves as a predictor of short-term recurrence.	Injury history as a criterion for returning to activity.
Pain-free walking	Warren P. et al. (2010) [45]	Pain-free walking time (those athletes who take more than 24 h to walk without pain are more likely to take more than 3 weeks to return to competition) is a significant independent predictor of return-to-competition time.	Time of pain when walking as a predictor of return to activity.
	Chu S. et al. (2016) [48]	Those athletes who took more than 1 day to walk without pain were more likely to take more than 3 weeks to return to activity.	Time of pain when walking as a predictor of return to activity.
Inflammation	Creighton D. et al. (2010) [27]	No inflammation or swelling upon return to activity.	Absence of inflammation as a criterion for returning to activity.

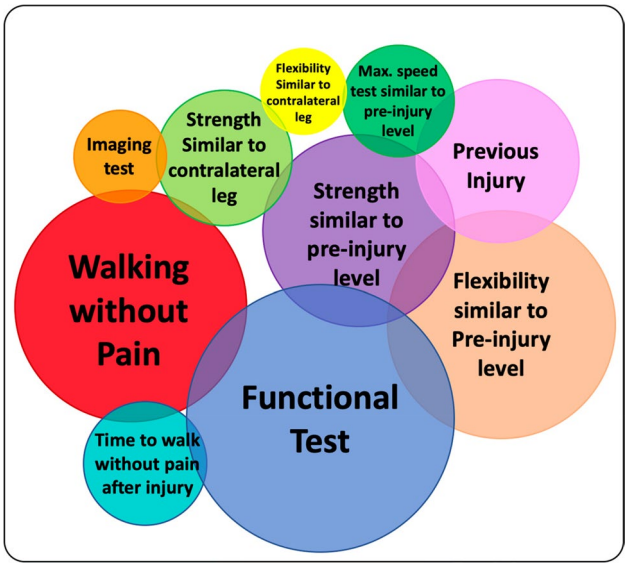


Fig. 3 Magnitude of the different criteria used for returning to activity according to the number of repetitions

This difference in determining complete recovery could be explained by the flexibility test selected, since the fact that a specific test for measuring flexibility in the hamstring muscles was not selected increased the relative error of the measurement, which may negatively influence the comparison between studies [77, 78]. Reliability has been determined in the existing literature in those tests that measure the flexibility of this muscle, both actively and passively, with the knee fully extended [79, 80] and with the hip flexed at 90° [79, 81]. The rest of the tests used must be analyzed in larger studies, where their sensitivity to differentiate between injury and non-injury values is determined, as well as their specificity to quantify the recovery process [39].

An increased risk of relapse has been shown when flexibility is incompletely recovered. In this regard, De Vos et al. [36] determined this risk with an adjusted odds ratio of 1.13 per degree deficit measured with the active knee extension test. When there is a sensation of instability during the active flexibility test, an increase of 1–2 weeks in the rehabilitation period is recommended [7, 39, 82]. On the other hand, it has been shown that a limitation in active knee flexibility [36] and hip abduction [83] increases the risk of muscle injury.

Strength

In the 13 articles that used the strength variable as a criterion for returning to activity, six different types of instruments were used. Despite this variability in the measurement instruments, it seems to be clear that the eccentric phase of the contraction of the hamstring muscles plays a more important role than the concentric phase in injuries of the hamstring muscle [17, 56, 84–87], with the largest number of muscular imbalances occurring in the eccentric phase [84]. In addition, it has been shown that an increase in eccentric strength decreases the risk of relapse [42, 85, 88]. When there is a decrease in eccentric strength, the capacity of the hamstring muscles to slow down the movement of the lower limb during the last phase of the running is reduced, which produces an increased risk of stretching and increases the risk of suffering an acute injury to said muscles [87, 88].

In relation to this last phase of the running, it has also been shown that a low ratio between hamstring muscles and quadriceps increases the risk of injury in this last phase of the running [87]. In professional American football players, it has been determined that a ratio of less than 0.50 increases the risk of injury in the hamstring muscles [89]; on the other hand, in professional soccer players, a ratio of 0.45–0.47 significantly increases the risk of relapse [84].

There is disagreement in whether determining the recovery of pre-injury values or comparing with the contralateral leg as a criterion for returning to activity. Thus,

it has been shown that a difference in ipsilateral or bilateral strength would be a consequence after an injury of the hamstring muscles [90, 91]. It has been demonstrated that the presence of an imbalance in muscle strength between both limbs increases the risk of muscle injury [87, 92], establishing a limit of 8–15% [55, 84, 88, 93].

Functional test measurement

Although the use of functional tests, especially those involving sport-specific movements, has been considered the gold standard, the existing evidence is scarce [22]. One of the reasons that could explain this aspect is the lack of technical equipment in rehabilitation centers for their implementation, such as GPS systems and oxygen meters [62]. This fact increases the need to implement an algorithm that integrates the performance of sport-specific movements and neuromuscular control exercises as a criterion for return to activity [58, 94].

The parameter that was most widely studied, with respect to the performance of functional tests, is related to the ability to run at maximum speed [25, 44, 46, 52, 59], since it has a direct relationship with the ability to develop horizontal force and the power developed [95]. In that sense, the moment of return to activity has been determined when the difference in the "60-m sprint time" test with respect to pre-injury levels is less than 2% [44]. In addition, it has been established that those athletes who have to reach a running speed greater than 24 km/h are those who have a greater probability of muscle injury, which could lead to taking this value as the minimum value to be reached without pain in these subjects at the time of return to activity [96]. Thus, in order to avoid relapses, the results seem to indicate that adaptive work by the rehabilitation team is necessary to reduce physical stress when performing a sprint [97].

Another parameter studied is related to the execution of movements typical of the sport at maximum speed. In this parameter, not only the data obtained in the tests performed are important, but also the subjective opinion of the injured person and the technical staff [98].

Other measurements

Regarding imaging tests, some authors are in favor of their use as a criterion for returning to activity [9, 27, 38, 52, 60], with the distance between the proximal area of the injury and the ischial tuberosity being the most relevant measure [39]. However, other authors report that their use is not necessary, especially in mild or moderate injuries [44, 65, 99], and that this information can be obtained by performing other tests that are less expensive [6]. What has been demonstrated is that performing imaging tests at the beginning of the injury serves as a predictive value for recovery time [33, 40, 100]. In any case, hamstring injuries have been classified into different grades (grades I, II and III) depending on the amount of pain, disability and loss of movement, which has been considered as a reflection of the amount of muscle and tendon fibers affected (from grade I, with minimal damage, to grade III, with a complete rupture) [101].

Another important parameter to take into account is the history of previous injuries in the hamstring muscles, since it can be considered as an independent prognostic indicator [102]. Those athletes with a history of previous injuries in this musculature will have a greater predisposition to suffering future injuries, which increases the need for preventive work in these patients [63, 103]. Age has also been related to an increase in the incidence of muscle injury [104]. We can add that the combination of a history of previous injury together with a delay of more than one day in returning to walking without pain increases by 93% the chances of returning to activity in more than 3 weeks [102].

Clinical implications

It is widely established that the diagnosis of a hamstring injury is based on the patient's previous injury history, a physical examination, and imaging tests [46]; however, there is still no consensus to determine the return to activity after this type of injury, which could generate controversy over which criteria to use, making it difficult to transfer this strategy to daily clinical practice [25]. This last aspect results in an excessively high relapse rate at present, which is coupled with a long rehabilitation period [87, 105]. It is important to highlight the importance of preventing relapses, since they are associated with a longer sick-leave period [5, 12, 20].

The absence of pain at the time of return to activity, as well as the recovery of the strength developed in said musculature together with the optimal performance of the functional tests, constitute the main criteria for a safe return to activity [23, 59]. The use of ultrasound imaging tests, better than the image obtained by magnetic resonance due to its difficult accessibility, serves as a support tool during the different phases of the rehabilitation period and not so much as a criterion for the return to activity [23].

According to the analyzed literature, preventive training of the hamstring muscles should focus on the eccentric phase [106, 107], which translates into an improvement in the work of said muscles during the sprint phase [3]. It is also important to achieve a correct activation of the hamstring muscles, in addition to reducing fatigue during muscle work and thus increasing their resistance [93, 108]. Finally, it is important to highlight the importance of flexibility work, since a correct length of the hamstring muscles will influence the correct development of muscle strength, in addition to providing an

optimal angle in the knee, which favors the action of the muscles [109].

Strengths and weaknesses

The main strength of the present review is that this is the first study to bring together all the existing literature on return-to-activity criteria in lower limb injuries, providing the clinician with a broad overview of the existing battery of tests and the functionality of their use. This review provides the clinician with a wide range of variables to analyze, both objective and subjective, including the presence of the latest imaging technology, allowing them to choose the appropriate tool according to the resources available and those most convenient for their patient. Furthermore, the inclusion of a wide variety of studies, in addition to the breadth of languages analyzed, means that this work compiles all the existing literature to date.

However, this study has certain limitations that should be taken into account when interpreting the results. For example, the lack of standardization when defining the return to activity in the literature makes it difficult to study, which is compounded by the lack and small number of quality studies. At the same time, although an attempt was made to cover practically all scientific articles published in the world, using the main databases and selecting documents in English, French, Portuguese, Italian and Spanish, there could be other potentially selectable studies that were not considered for the reasons mentioned above. In this sense, although the most important databases that collect the most relevant literature published in health sciences have been consulted, there could be others that could contain other studies that were not included in this review.

Furthermore, it is crucial to acknowledge the inherent heterogeneity in the study designs included in this review, which encompasses randomized controlled trials, observational longitudinal studies, and systematic reviews. This heterogeneity complicates the direct comparison of outcomes. Additionally, there is a notable inconsistency in the methodological quality across the analyzed literature. As demonstrated in our quality assessments, PEDro scores ranged from 3 to 6 out of 10, STROBE scores varied widely from 8 to 19 out of 22, and PRISMA scores showed significant dispersion from 2 to 17 out of 27. This variability underscores an urgent need for future research in this field to significantly improve overall methodology, transparency, and reporting standards. To establish highly reliable and universal return-to-play criteria, subsequent studies must prioritize the standardization of measurement instruments and explicitly detail functional testing protocols.

Conclusion

The systematic analysis of the current scientific literature demonstrates that clinical decision-making for the return to play following a hamstring muscle injury has shifted toward a multifactorial and criteria-based paradigm. A safe return is no longer determined by a single isolated marker or a rigid timeline, but rather by a synergistic combination of biological healing, clinical stability, and functional readiness. The evidence highlights the objective measurement of physical qualities as a primary pillar, establishing a clear consensus on the necessity of achieving biomechanical symmetry. Specifically, muscle strength and flexibility must be restored to match the uninjured limb, typically accepting a maximum deficit of 10% as a safety threshold to minimize the risk of relapse. While advanced diagnostic tools like Magnetic Resonance Imaging are essential for characterizing anatomical severity and tissue healing, findings suggest that imaging results must be integrated with functional performance, as structural signals of injury can often persist even when an athlete is clinically ready for competition.

The transition to high-level play is increasingly guided by dynamic, sport-specific criteria that bridge the gap between clinical rehabilitation and competitive demands. Successful discharge requires the completion of functional batteries that include maximum-speed sprints, acceleration-deceleration patterns, and sport-specific movements performed without pain or instability. This progression is underpinned by structured exercise protocols that prioritize a gradual increase in mechanical load and respect for biological healing times. Despite these advances, the literature reveals a need for greater standardization in how pain thresholds and specific measurement instruments are reported to ensure more reproducible decision-making. In conclusion, a successful return to sport following a hamstring injury requires a transition from time-based protocols to a rigorous, criteria-based progression that validates both objective physical metrics and the athlete's ability to meet the high-intensity functional demands of their specific sport.

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Authors' contributions
JJBBR was responsible for writing the article. DPC and MGS were responsible for analyzing the existing literature. JGR, MRM, and JCCS were responsible for developing the article's design and layout.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

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Not applicable.

Consent for publication
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Competing interests
The authors declare no competing interests.

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